

Potential Outcomes

Causal Model: $Y_i = \alpha_0 + \alpha_1 D_i + u_i$, $Y_i(0) = \alpha_0$
 $\Rightarrow Y_i(0) = D_i Y_i(0) + (1-D_i) Y_i(0)$

ATE = $E[Y_i(1) - Y_i(0)] = \alpha_1$
ATT = $E[Y_i(1) - Y_i(0) | D_i=0]$

ATE with imperfect compliance is identified by N only if the treatment effect is constant (homogeneous) across individuals: $Y_i = Y_i(0) + Y_i(1) + A_i$

Causal Model w/ constant TE:
 $Y_i = Y_i(0) + D_i (Y_i(1) - Y_i(0)) = \alpha_0 + A_i D_i + u_i$

where $A_i \equiv Y_i(0) - Y_i(1)$, $A_i = E[Y_i(0)]$

$\hat{\beta}_0 = \frac{1}{N} \sum_{i=1}^N Y_i(0) - \frac{1}{N} \sum_{i=1}^N Y_i(1)$

$= \frac{1}{N} \sum_{i=1}^N (Y_i(0) - Y_i(1)) = \frac{1}{N} \sum_{i=1}^N Y_i(0) - \frac{1}{N} \sum_{i=1}^N Y_i(1)$

$= E[Y_i(0) | D_i=1] - E[Y_i(0) | D_i=0]$ by LIE

$= \alpha_1$ selection bias

3.1 Lec 7 (IV-LATE) - α_2 in D_1

$Y_i, D_i \sim i$'s person's potential outcome when $D_i \in \{0,1\}$

$Z_i \sim$ instrument $\{1\}$ D

1) Random assignment of Z_i : $D_i(0), D_i(1)$ & Z_i

2) Exclusion Restriction: $Y_i(D, z) = Y_i(D)$

3) Monotonicity: $D_i(1) \geq D_i(0) \Rightarrow D_i(1)=1$ if $D_i(0)=0$

First-Stage: $D_i = \pi_0 + \pi_1 Z_i + \epsilon_i$
 Reduced-Form: $Y_i = \delta_0 + \delta_1 Z_i + v_i$

	$Z_i=0$	$Z_i=1$
$D_i(0)$	$Y_i(D,0)$	$D_i(1)$
$D_i(1)$	$Y_i(D,0)$	$Y_i(D,1)$
AT	1	1
NT	0	0
Compliers	0	1
Defiers	1	0

g_i : denotes 4 groups; Defiers go away with monotonicity ($D_i(0) > D_i(1)$)

How to derive expressions for population estimates of first stage coefficient π_1 :

$\pi_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0]$

$\phi[E[D_i | Z_i=1]] = E[D_i(0) | Z_i=1] = E[D_i(0)]$; by LIE we get:

$E[D_i | Z_i=1] = E[E[D_i | Z_i=1, g]]$

$= \sum_g \Pr(g) \cdot E[D_i(0) | Z_i=1, g]$

$= \sum_g \Pr(g) \cdot E[D_i(0) | g]$ by random assignment independence $Z_i \perp (D_i(0), D_i(1))$ or $g^* \perp g$

Since $D_i(1)=1$ for AT & Compliers:

$\Rightarrow E[D_i | Z_i=1] = \sum_g \Pr(g) \cdot E[D_i(0) | g] = \Pr(AT) + \Pr(Comp)$

Similarly since $D_i(0)=1$ for AT & Defiers:

$\Rightarrow E[D_i | Z_i=0] = \sum_g \Pr(g) \cdot E[D_i(0) | g] = \Pr(AT) + \Pr(Defier)$

So our first stage coefficient (π_1):

$\pi_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0] = \Pr(AT) + \Pr(Comp) - [\Pr(AT) + \Pr(Defier)]$

$\Rightarrow \pi_1 = \Pr(Comp) - \Pr(Defier)$

$\Rightarrow$ If more compliers than defiers $\Rightarrow \pi_1 > 0$

$\Rightarrow \pi_1 = \Pr(Comp)$ if we have monotonicity (no defiers)

Head Start Program ~ IV-LATE framework

- 1) Relevance: $\pi_1 \neq 0$
- 2) Independence/Exclusion Restriction: $Z_i \perp (Y_i(0), Y_i(1))$ where Z_i only affects Y_i through D_i s.t. $E[Z_i | (Y_i(0), Y_i(1))] = 0$
- 3) Constant Treatment Effect (CATE): $Y_i = Y_i(0) + Y_i(1) = \alpha_1$ *

Causal Model: $Y_i = \alpha_0 + \alpha_1 D_i + u_i$

1st Stage: $D_i = \pi_0 + \pi_1 Z_i + \eta_i$ $\{ \pi_0 = E[D_i | Z_i=0] \}$
 $\pi_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0]$

2nd Stage: Plug 1st stage into causal:
 $Y_i = \alpha_0 + \alpha_1 (\pi_0 + \pi_1 Z_i + \eta_i) + u_i$
 $= \alpha_0 + \alpha_1 \pi_0 + \alpha_1 \pi_1 Z_i + \alpha_1 \eta_i + u_i$

Since $E[Z_i | (Y_i(0), Y_i(1))] = 0$ by random assignment/exclusion restriction we can estimate a reduced-form regression of Y_i on $(1, Z_i)$ to get

Reduced Form: $Y_i = \delta_0 + \delta_1 Z_i + \epsilon_i$

$\Rightarrow \hat{\beta}_1 = \frac{\delta_1}{\pi_1}$

$\hat{\pi}_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0] = \frac{1}{N} \sum_{i=1}^N D_i - \frac{1}{N} \sum_{i=1}^N D_i = \bar{D}_1 - \bar{D}_0$

$\hat{\pi}_1 = \frac{1}{N} \sum_{i=1}^N D_i = \bar{D}_1$

Under (1-3*): $\frac{\delta_1}{\pi_1} \xrightarrow{p} \alpha_1 = ATE$

$\hat{\pi}_1$ represents compliance: the effect of HS offer on the actual enrollment in Head Start or compliance w/ the randomly assigned treatment Z_i (in IV-LATE, $\hat{\pi}_1$ ~ estimator of the fraction of compliers who would choose $D_i=0$ (not taking up the HS seat) if they were assigned to control group without an initial offer, and who would take up $D_i=1$ if they were assigned to the treatment group).

Under (1-2), if we have monotonicity ($D_i(1) \geq D_i(0) \Rightarrow$ no defiers)

$\Rightarrow \frac{\delta_1}{\pi_1} = E[Y_i(0) - Y_i(1) | Compliers] = LATE$

• (NT) Share of families who didn't take up HS seat despite being in treatment group $= (1 - 0.857) = 0.143 \sim E[D_i=0 | Z_i=1]$

• (AT) share of families who took HS seat despite being in control group $= \bar{D}_0 = \hat{\pi}_0 = 0.153$

• (Compliers) share of families who 'complied' w/ random assignment: π_1

* have no way of knowing which family is AT vs Complier

• From reduced form ($Y_i = \delta_0 + \delta_1 Z_i + \epsilon_i$) $\Rightarrow \hat{\delta}_1 = \bar{Y}_1 - \bar{Y}_0$ where $\hat{\delta}_1$ ~ intent to treat (ITT): the impact of HS offer (winning the lottery) on the outcome of kid. See below:

$N_1 = \sum_{i=1}^N Z_i$ (# units assigned to treatment) $\cdot \bar{D}_1 = \frac{1}{N} \sum_{i=1}^N D_i$ (avg. of D among lottery winners)

$N_0 = \sum_{i=1}^N (1-Z_i)$ (# units assigned to control) $\cdot \bar{Y}_1 = \frac{1}{N} \sum_{i=1}^N Y_i$ (avg. outcome of lottery winners)

$\hat{\delta}_0 = \bar{Y}_0 = \frac{1}{N} \sum_{i=1}^N Y_i \sim$ avg. outcome for losers in lottery (winners control group)

$\hat{\delta}_1 = \bar{Y}_1 - \bar{Y}_0 = (\frac{1}{N} \sum_{i=1}^N Y_i) - (\frac{1}{N} \sum_{i=1}^N Y_i)$

Second Stage (2SLS): $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \hat{D}_i$

$\hat{\beta}_1 = \frac{\hat{\delta}_1}{\hat{\pi}_1} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{D}_1 - \bar{D}_0} = E[Y_i(1) - Y_i(0) | Compliers] = LATE \rightarrow \alpha_1$ CATE

* avg. causal effect of HS enrollment among compliers whose treatment (status) is induced by the instrument (lottery).

Smoking (Counseling IV) with Two Mutually-Exclusive Behaviors

Empirical story: Z_i (0,1) randomized counseling offer (Z_i treated).

- Outcome Y is latent behavior.
- Two binary mutually exclusive behaviors at delivery: $D_i = 1$ if the mother does not quit but changes behavior (e.g., low-intensity smoking / NRT only).
- Potential outcomes $Y_i(D_i, Z_i)$ are any given Z_i and any given D_i (e.g., $Y_i(0,0)$ = no NRT, no quit).
- Random assignment: $(D_i(0), D_i(1), D_i(0), D_i(1), Y_i(0,0), Y_i(0,1), Y_i(1,0), Y_i(1,1))$ ~ i.i.d.
- Smoking given instrumentally exogenous for the uptake of treatment. For any uptake: $T(1) \geq T(0)$. Optionally, monotonicity

Principal strata for $(D_i(0), D_i(1)) = (D_i(1), D_i(0))$:

NT: $(0,0) \rightarrow (0,0)$
 Always NT: $(1,0) \rightarrow (1,0)$
 Always D1: $(0,1) \rightarrow (0,1)$
 Always D2: $(1,1) \rightarrow (1,1)$
 If no NRT, D1-defiers: $(1,0) \rightarrow (0,0)$
 If no NRT, D2-defiers: $(0,1) \rightarrow (0,1)$

First-stages (what Z shifts).

$\pi^{(1)} = E[D_1 | Z=1] - E[D_1 | Z=0] = \Pr(D_1=comp) - \Pr(D_1=def)$

$\pi^{(2)} = E[D_2 | Z=1] - E[D_2 | Z=0] = \Pr(D_2=comp) - \Pr(D_2=def)$

$\pi^{(T)} = E[T | Z=1] - E[T | Z=0] = \pi^{(1)} + \pi^{(2)}$

Reduced form for birthweight.

$\delta = E[Y | Z=1] - E[Y | Z=0] = E[Y(1,0) - Y(0,0) | D_1=comp] \Pr(D_1=comp) + E[Y(0,1) - Y(0,0) | D_2=comp] \Pr(D_2=comp)$

$- E[Y(1,0) - Y(0,0) | D_1=def] \Pr(D_1=def) - E[Y(0,1) - Y(0,0) | D_2=def] \Pr(D_2=def)$

Which Wald ratio equals which LATE? (State the extra condition you use.)

$\frac{\delta}{\pi^{(1)}} = \begin{cases} E[Y(1,0) - Y(0,0) | D_1=comp], & \text{if (i) } D_1\text{-monotonicity and (ii) either } \pi^{(2)} = 0 \text{ or } Y(0,1) = Y(0,0); \\ \text{signed mix of types,} & \text{otherwise.} \end{cases}$

$\frac{\delta}{\pi^{(2)}} = \begin{cases} E[Y(0,1) - Y(0,0) | D_2=comp], & \text{if (i) } D_2\text{-monotonicity and (ii) either } \pi^{(1)} = 0 \text{ or } Y(1,0) = Y(0,0); \\ \text{signed mix of types,} & \text{otherwise.} \end{cases}$

$\frac{\delta}{\pi^{(T)}} = \begin{cases} E[\tau_1 | D_1=comp] \Pr(D_1=comp) + E[\tau_2 | D_2=comp] \Pr(D_2=comp), & \text{if T-monotonicity;} \\ \Pr(D_1=comp) + \Pr(D_2=comp) & \text{otherwise,} \end{cases}$

where $\tau_1 \equiv Y(1,0) - Y(0,0)$ is the birthweight gain from quitting, and $\tau_2 \equiv Y(0,1) - Y(0,0)$ is the gain from the non-quit behavior (e.g., low-intensity/NRT).

Fast "goofy" formulas to recover complier means (smoking context).

$\frac{\delta \cdot Y(1,0)}{\pi^{(1)}} = E[Y(1,0) | D_1=comp],$

$\frac{\delta \cdot X(1,0)}{\pi^{(1)}} = E[X | D_1=comp],$

$\frac{\delta \cdot X(1,-D_2)}{\pi^{(1)}} = E[X(1,-D_2) | D_1=comp],$

$\frac{\delta \cdot X(0,D_2)}{\pi^{(2)}} = E[X | D_2=comp],$

for any covariate X (e.g., maternal age). Analogous formulas hold swapping D_1/D_2 .

1. Draw the table: rows = NT / always-D1 / always-D2 / D1-comp / D2-comp (and defiers if allowed); columns Z=0,1; fill (D_1, D_2) and realized Y .
2. Compute $\pi^{(1)}, \pi^{(2)}, \pi^{(T)}$ and interpret (net shares moved by counseling).
3. Write the reduced form δ by differencing the two Z columns.
4. Pick the relevant Wald ratio and state the extra condition that kills the "other-behavior" term.
5. (Optional) Use goofy products to back out $E[Y(1,0)]$, $E[Y(0,0)]$ or complier characteristics.

2) Derive an expression for δ_1 in the reduced form:

$\delta_1 = E[Y_i | Z_i=1] - E[Y_i | Z_i=0]$

By exclusion restriction, conditional on Z_i , realized $D_i = D_i(Z_i)$ and realized outcome $Y_i = Y_i(D_i(Z_i))$. Using LIE:

$\star E[Y_i | Z_i=0] = E_g[E[Y_i | Z_i=0, g]] = E_g[E[Y_i(0,0) | Z_i=0, g]]$

$= E_g[E[Y_i(D_i(0), 0) | g]]$

Where $Z_i=0$ is dropped in the last equation b/c treatment is randomly assigned ($Z_i \perp (D_i(0), D_i(1), Y_i(0), Y_i(1)) \Rightarrow E_g[E[Y_i(D_i(0), 0) | g]] = \Pr(AT) \cdot E[Y_i(0) | AT]$

$+ \Pr(NT) \cdot E[Y_i(0) | NT] + \Pr(Comp) \cdot E[Y_i(0) | Comp] + \Pr(Defier) \cdot E[Y_i(0) | Defier]$

Similarly, conditional on being assigned to the treatment group $Z_i=1$

$\star E[Y_i | Z_i=1] = E_g[E[Y_i(D_i(1), 1) | g]] = \Pr(AT) \cdot E[Y_i(1) | AT] + \Pr(NT) \cdot E[Y_i(1) | NT]$

$+ \Pr(Comp) \cdot E[Y_i(1) | Comp] + \Pr(Defier) \cdot E[Y_i(1) | Defier]$

AT & NT have $Y_i(D_i(0)) = Y_i(D_i(1))$ as $D_i(0)=D_i(1)$, so the reduced form coefficient

$\delta_1 = E[Y_i | Z_i=1] - E[Y_i | Z_i=0]$

$\hat{\delta}_1 = \Pr(Comp) \cdot E[Y_i(1) - Y_i(0) | Comp] + \Pr(Defier) \cdot E[Y_i(1) - Y_i(0) | Defier]$ *

3) What is $\frac{\delta_1}{\pi_1}$ when $E[Y_i(1) - Y_i(0) | Compilers] = E[Y_i(1) - Y_i(0) | Defiers]?$

$\Rightarrow \frac{\delta_1}{\pi_1} = \frac{\Pr(Comp) \cdot E[Y_i(1) - Y_i(0) | Comp] + \Pr(Defier) \cdot E[Y_i(1) - Y_i(0) | Defier]}{\Pr(Comp) - \Pr(Defier)} = E[Y_i(1) - Y_i(0) | Comp]$

That is, the IV estimator would again give us the LATE of compliers!

Lec 8 Panel Data - Twins Study

- Set-up**
- Each family has a pair of identical twins
 - i index a family; $i \in \{1, 2, \dots\}$ - denote each twin
 - Goal: Use (i, i) panel data to estimate the returns to educ.
 - $Y_{it} \sim$ log earnings on educ S_{it} + controls
- Model Specification**
- $Y_{it} = \beta_0^{FE} + \beta_1^{FE} S_{it} + \alpha_i + \epsilon_{it}$
 - $Y_{it} = \beta_0^{CF} + \beta_1^{CF} S_{it} + \bar{S}_i + u_{it}$ **control fcn. w/ family mean**
 - $Y_{it} = \beta_0^{CF} + \beta_1^{CF} S_{it} + \delta_1^{CF} S_{it-1} + v_{it}$
- (1) controls for family for fixed effects (controlling for dummies of each family)
- (2) represents control function approach w/ a control $\bar{S}_i = \frac{S_{it} + S_{it-1}}{2}$, the mean educ of twins in the same family.
- (3) - alternative control function that controls for S_{it-1} which is the educ of the identical twin instead of $\bar{S}_i$. for example, for twin $i=1$ in family (i), $S_{it-1} = S_{it}$ -> the educ of his/her identical twin in the family.

- $\alpha_i \sim$ "everything about family (i) that doesn't change across the 2 twins - genes, parents, background, neighborhood, preferences, etc."
- $\Rightarrow$ including α_i is equivalent to demeaning within family (or first differencing with 2 twins): $(Y_{it} - Y_{it-1}) = \beta_1^{FE} (S_{it} - S_{it-1}) + (\epsilon_{it} - \epsilon_{it-1})$ so that β_1^{FE} is identified only from differences between the twins in the same family - not across families.

To read β_1^{FE} causally, we need **strict exogeneity** (that the residual ϵ_{it} at $t=0$ is mean independent of the educ. attainment of the twins and the family fixed effect) $\rightarrow E[\epsilon_{it} | S_{it}, S_{it-1}, \alpha_i] = 0$

Under strict exogeneity: $\hat{\beta}_1^{FE} = \hat{\beta}_1^{CF}$ b/c (1) can be rewritten as:
 $Y_{it} = \beta_0^{FE} + \beta_1^{FE} S_{it} + \sum_{j=2}^J D_{ij} \alpha_j + \epsilon_{it}$, which controls for dummies of all fambes except $C(1)$ one as we controlled for constant (alternatively, drop 1 and control for dummies for all families from 1 to N): $D_{ij} = 1$ if $j=i$. Applying Frisch-Waugh to (1) under strict exogeneity ($E[\epsilon_{it} | S_{it}, S_{it-1}, \alpha_i] = 0$):

$\hat{\beta}_1^{FE} = \frac{\text{cov}(Y_{it}, S_{it})}{\text{var}(S_{it})}$ where $\bar{S}_i = S_{it} - (\bar{\alpha}_i + \sum_{j=2}^J \bar{\alpha}_j D_{ij})$

The aux. reg. of S_{it} on $(1, D_{i2}, D_{i3}, \dots, D_{iJ})$ is the mean for each family: $\bar{S}_i = \bar{S}_{i0} \quad \forall j \geq 2, \bar{S}_j = \bar{S}_{i0} + \bar{\alpha}_j$ and therefore $\bar{S}_{it} = S_{it} - \bar{S}_i$

Now we use FW to control function model (2):
 $\hat{\beta}_1^{CF} = \frac{\text{cov}(Y_{it}, S_{it}^{CF})}{\text{var}(S_{it}^{CF})}$, where $S_{it}^{CF} = S_{it} - (\bar{\alpha}_i + \sum_{j=2}^J \bar{\alpha}_j D_{ij})$

The aux. model regresses S_{it} on $(1, \bar{S}_i)$ and as shown in Lec 7, we have $S_{it} = 0, \hat{\alpha}_i = 1 \Rightarrow \hat{S}_{it}^{CF} = S_{it} - \bar{S}_i = \bar{S}_i \Rightarrow \hat{\beta}_1^{CF} = \hat{\beta}_1^{FE}$

If strict exogeneity fails $\Rightarrow \beta_1^{FE} = \beta_1 + \frac{\text{cov}(\epsilon_{it}, S_{it-1})}{\text{var}(S_{it} - \bar{S}_i)}$

so both FE & (2) w/ $\bar{S}_i$ converge to the same biased pseudo-parameter (the partial slope from regressing Y on S after partialling out family means) $\neq$ causal effect. The bias term's sign depends on how the within-family schooling difference correlates w/ the twin-specific error (i.e. twin-specific ability $\Rightarrow$ upward bias; classical measurement error in $S \Rightarrow$ attenuation).

Under Strict Exogeneity: $\hat{\beta}_1^{CF} = \hat{\beta}_1^{FE} - \hat{\delta}_2^{CF}$ b/c we plug $\bar{S}_i = \frac{S_{it} + S_{it-1}}{2}$

into (2): $\hat{\beta}_1 = \hat{\beta}_1^{CF} + \hat{\beta}_1^{FE} \cdot \left(\frac{S_{it} + S_{it-1}}{2} \right) \hat{\gamma} = \hat{\beta}_1^{CF} + \left(\hat{\beta}_1^{FE} \cdot \frac{1}{2} \right) S_{it-1} + \frac{1}{2} S_{it-1}$

Given that the orthogonality condition is solved by the OLS estimators, $(Y_{it} - \bar{Y}_{it}) \perp (1, S_{it}, \bar{S}_i)$. After re-arranging the terms, we obtain in model (3):
 $\hat{\delta}_1^{CF} = \hat{\beta}_1^{CF} + \frac{1}{2} \quad \& \quad \hat{\delta}_2^{CF} = \frac{1}{2} \Rightarrow \hat{\delta}_1^{CF} - \hat{\delta}_2^{CF} = \hat{\beta}_1^{CF}$

Alternative Estimates of Return to Education Using Twins Sample					
$Y_{it} = \log(\text{wages})$ $\Rightarrow$ effect on wages: $\approx 100 \cdot (\exp(\beta) - 1)$ $\approx 100 \cdot (e^{\beta} - 1)$	(1) OLS	(2) OLS with Family Dummies	(3) OLS with Mean Educ. Of Pair	(4) OLS with Educ. Of Sibling	(5) OLS with Educ. Of Father
Own Education	0.0802 $\hat{\beta}_1$ (0.0071)	0.0678 $\hat{\beta}_1^{FE}$ (0.0131)	0.0678 $\hat{\beta}_1^{CF}$ (0.0197)	0.0749 $\hat{\beta}_1^{CF}$ (0.0106)	0.0843 $\hat{\beta}_1$ (0.0075)
Mean Educ. of Pair	--	--	0.0143 $\hat{\gamma}$ (0.0212)	--	--
Sibling Education	--	--	b/c $\hat{\beta}_1^{FE} = \hat{\beta}_1^{CF}$	0.0072 $\hat{\delta}_2^{CF}$ (0.0106)	--
Father's Education	--	--	$\hat{\delta}_2^{CF} = \hat{\beta}_1^{FE} - \hat{\beta}_1^{CF}$	--	-0.0097 $\hat{\delta}_1^{CF}$ (0.0052)

Key Results / Identities

$\hat{\beta}_1^{FE} = \hat{\beta}_1^{CF} + \hat{\delta}_2^{CF}$

$\hat{\beta}_1^{FE} = \hat{\beta}_1^{CF} + \hat{\delta}_2^{CF}$

$\hat{\beta}_1^{FE} = \hat{\beta}_1^{CF} + \hat{\delta}_2^{CF}$

In this question we will use potential outcomes to interpret the difference-in-differences model. Consider the classic Card and Krueger (1994) minimum wage problem. Let i denote a restaurant that either be in NJ or in PA (but not both - ignore chains like McDonalds that have locations in both states). $t = 1$

denotes the period before NJ raised minimum wage, and $t = 2$ denotes the period after the change. As before, define $NJ_t = 1$ if restaurant i is in NJ, and $Post_t = 1$ if $t = 2$ (after the minimum wage increase).

Define the treatment variable as $D_{it} = NJ_t \times Post_t$, which equals to 1 if the restaurant i is in NJ, and $t = 2$. The outcome of interest is full-time employment Y_{it} , and we want to estimate the difference-in-differences regression:

$$Y_{it} = \beta_0 + \beta_1 NJ_t + \beta_2 Post_t + \beta_3 D_{it} + \epsilon_{it}$$

We can write the outcome as realized $Y_{it} = Y_{it}(D_{it})$. That is, $Y_{it}(0)$ is the potential full-time employment when there is no change in minimum wage, and $Y_{it}(1)$ is the potential outcome when the minimum wage is increased.

- (a) Can you estimate $E[Y_{it}(0)|NJ]$ and $E[Y_{it}(0)|PA]$ in the data? Explain why or why not.
- Yes. At $t = 1$, $D_{it} \equiv 0$, and therefore the realized outcome $Y_{it} = Y_{it}(0)$ for each restaurant. We can get consistent estimates by computing the sample means in each state:

$$\frac{1}{N_1} \sum_{i \in NJ} Y_{it} \xrightarrow{P} E[Y_{it}(0)|NJ]$$

$$\frac{1}{N_2} \sum_{i \in PA} Y_{it} \xrightarrow{P} E[Y_{it}(0)|PA]$$

in which N_1 = num. restaurants in NJ in our sample, and N_2 = ... in PA.

- (b) Can you estimate $E[Y_{it}(1)|NJ]$ and $E[Y_{it}(1)|PA]$ in the data? Explain why or why not.
- At $t = 2$, for those in NJ, $D_{it} = 1$, and we observe $Y_{it} = Y_{it}(D_{it}) = Y_{it}(1)$. Therefore we can estimate $E[Y_{it}(1)|NJ]$ by the sample mean of Y_{it} among NJ restaurants at $t = 2$. In contrast, for those in PA, $D_{it} = 0$ and $Y_{it} = Y_{it}(0)$, and we cannot estimate $E[Y_{it}(1)|PA]$.

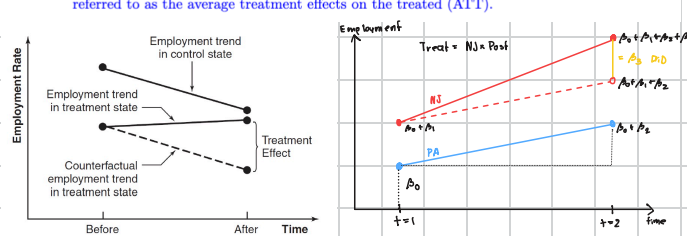
- (c) Use potential outcomes $\{Y_{it}(0), Y_{it}(1)\}$ to express the parallel trend assumption. Under the parallel trend assumption, what does β_3 represent?
- The parallel trend assumption says in the absence of the min-wage change in NJ, the change in employment in NJ would have been the same as the change in employment in PA. In potential outcomes framework, it can be stated as

$$E[Y_{it}(0) - Y_{it}(1)|NJ] = E[Y_{it}(0) - Y_{it}(1)|PA]$$

Under the parallel trend assumption, the difference-in-difference coefficient

$$\begin{aligned} \beta_3 &= E[Y_{it}(1) - Y_{it}(0)|NJ] - E[Y_{it}(0) - Y_{it}(1)|PA] \\ &= E[Y_{it}(1) - Y_{it}(0)|NJ] - E[Y_{it}(0) - Y_{it}(1)|NJ] \\ &= E[Y_{it}(1) - Y_{it}(0)|NJ] \end{aligned}$$

represents the causal effect of the minimum wage change on employment in NJ. It's also referred to as the average treatment effects on the treated (ATT).



- (1) $Y_{it} = \beta_0 + \beta_1 S_{it} + u_{it}$ (OLS)
 $\beta_1 = 0.0802 \sim$ A 1-year increase in own schooling is associated w/ about $(e^{0.0802} - 1) = 8.3\%$ higher wages. Not causal - it mixes within & between-family variation and can reflect family ability/background.
- (2) $Y_{it} = \beta_0^{FE} + \beta_1^{FE} S_{it} + \alpha_i + \epsilon_{it}$ (OLS w/ Family Dummies (Fixed Effects))
 $\hat{\beta}_1^{FE} = 0.0678 \sim$ Within family return to schooling. Comparing twins in the same family, an extra year of schooling raises wages by $\approx 6.8\%$, assuming strict exogeneity ($E[\epsilon_{it} | S_{it}, S_{it-1}, \alpha_i] = 0$; Family - constant factors are soaked up).
- (3) $Y_{it} = \beta_0^{CF} + \beta_1^{CF} S_{it} + \bar{S}_i + u_{it}$ (OLS w/ mean educ. of pair of twins (control function))
 $\hat{\beta}_1^{CF} = 0.0678 \sim$ The fixed effects slope: the within-family return of +1 year of educ $\approx 6.8\%$.
 $\hat{\gamma} = 0.0143 \sim$ The between-family correlation: Conditional on your own schooling, being from a family with 1 additional year of average schooling is associated with $\approx 1.4\%$ higher wages (not causal; it soaks up shared family influences).
- (4) $Y_{it} = \beta_0^{CF} + \beta_1^{CF} S_{it} + \hat{\delta}_2^{CF} S_{it-1} + v_{it}$ (OLS w/ Educ. of sibling (other-twin regressor))
 $\hat{\delta}_1^{CF} = 0.0749 \sim$ (Own educ): conditional on sibling's schooling, a 1-year increase in a twin's schooling is associated with $\approx 7.4\%$ higher wages.
 $\hat{\delta}_2^{CF} = 0.0072 \Rightarrow \hat{\beta}_1^{FE} - \hat{\delta}_2^{CF} = \hat{\beta}_1^{CF} = 0.0749 - 0.0072 = 0.0678$ & $\hat{\delta}_2^{CF} = \frac{1}{2}$. This calc. gives you FE what dummies by subtracting sibling's schooling from "own".
- (5) $Y_{it} = \beta_0 + \beta_1 S_{it} + \beta_2 F_i + \epsilon_{it}$ (OLS w/ Educ. of Father (Proxy for family background))
 $\hat{\beta}_1 = 0.0843 \sim$ Conditional on Father's schooling, a 1-year increase in your schooling is associated with $\approx 8.4\%$ higher wages. This does not replicate the within-family correlation: a single family proxy (Father's schooling) cannot fully absorb all family factors of the Father's schooling coefficient in a correlation, not causal; sign and size reflect how Father's schooling co-moves with both the twin's own schooling and unobservables.